will provide benefits to market participants that previously only relied on the SIP and choose to receive the new core data elements if the fees are lower as part of consolidated market data than fees for equivalent data today.[1924] Expanding core data will reduce information asymmetries between these market participants and market participants that subscribe to proprietary DOB feeds.[1925] A reduction in information asymmetry may, in turn, enhance market efficiency and price discovery if it leads to information that was previously only contained in proprietary DOB feeds being impounded into prices quicker.[1926] The additional information contained in expanded core data will also allow these market participants to improve order routing and will help facilitate best execution, which may reduce their transaction costs.[1927] The additional content of expanded core data could make consolidated market data a reasonable alternative to

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1924 See infra Section V.C.2(b)(i).

1925 Commenters agreed that the expansion of core data would reduce information asymmetries. See, e.g., BestEx Research Letter at 2; Better Markets Letter at 2–3; BlackRock Letter at 2; Capital Group Letter at 2. See infra note 2404 and accompanying text for a discussion of commenter stating that allowing competing consolidators to offer customized products containing subsets of expanded core data would increase information asymmetries.

1926 Commenters agreed that the expansion of core data would improve market efficiency and price discovery. See, e.g., Better Markets Letter at 2–3; ICI Letter at 5.

1927 Commenters agreed that the additional information in core data would facilitate best execution. See, e.g., Clearpool Letter at 11; DOJ Letter at 4; IntelligentCross Letter at 2; SIMFA Letter at 3–4.

exchange proprietary data feeds for some market participants,[1928] potentially lowering their costs.[1929]

One commenter stated that it is unclear whether the expanded content of core data would be useful to any set of investors and that the Proposing Release did not provide any analysis on this point.[1930] This commenter questioned whether there would be demand for the expanded content of core data, stating that it would simultaneously provide "too much and too little to be optimal for anyone – too much data for the retail investor and too little for sophisticated traders."[1931] This commenter also stated that expanding the content of core data would provide no real benefits because all of the information is already available to everyone who needs it.[1932] The Commission disagrees with this commenter and believes there would be demand for the expanded content of core data. Although the Commission did not quantify the number of market participants that would subscribe to the expanded content of core data, the Commission did provide a qualitative analysis of how certain market participants might subscribe to and could

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1928 Commenters agreed that the expanded content of core data could reduce some market participants' dependence on proprietary data feeds. See, e.g., Clearpool Letter at 11; BlackRock Letter at 2; DOJ Letter at 4.

1929 See infra Section V.C.2(b) (discussing potential fees for consolidated market data). Commenters agreed the expanded content of core data could lower costs for some market participants who currently subscribe to proprietary DOB feeds and switch to consolidated market data. See, e.g., Virtu Letter at 5.

1930 See Nasdaq Letter IV at 33, 38 ("the Proposed Rule replaces "only pay for what you need" with a feed that is simultaneously providing too much and too little to be optimal for anyone – too much data for the retail investor and too little for sophisticated traders").

1931 See Nasdaq Letter IV at 38.

1932 See Nasdaq Letter IV at 34.

benefit from the expanded content of core data.[1933] Although expanded core data will not contain all of the data contained in proprietary DOB feeds, the Commission believes that it will contain data that will be useful for market participants.[1934] For example, although the DOB data contained in expanded core data will only contain five levels of depth, the Commission believes, and commenters agree, that including five levels of depth in expanded core data will provide a benefit to market participants, including allowing them to improve their order routing.[1935] The Commission believes that there are market participants who would subscribe to proprietary DOB feeds, but do not currently do so because of the cost.[1936] Because the Commission anticipates that the total fees for a consolidated market data product containing all the elements of expanded core data are likely to be less expensive than equivalent proprietary data feeds,[1937] the Commission believes that there would be demand from these market participants for a

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1933 See Proposing Release, 85 FR at Section VI.C.1.

1934 For example, expanded core data will not contain complete order-by-order information or full depth of book information.

1935 See infra Section V.C.1(c)(ii). Commenters agreed that five levels of depth is sufficient for many market participants. See, e.g., State Street Letter at 2–3; Capital Group Letter at 3; Fidelity at 4.

1936 See supra note 28 (discussing commenters' views that the cost of proprietary DOB products currently inhibits the purchase of, and the widespread dissemination of, the data elements that will be contained in expanded core data). See also Roundtable Day One Transcript at 128–29 (Mark Skalabrin, Redline Trading Solutions) (stating that some customers do not purchase exchange proprietary DOB products because of the cost, explaining "we sell to various customers, leading firms that have lots of money and really imbed this technology, but also to startup brokers and small firms trying to integrate in the market. And not all of them use direct feeds. And it was mentioned before that some people just don't buy the direct feeds. Some people can do without it. And we deal with them in that decision process. It's not a mystery why they don't use the direct feeds; it's solely cost.").

1937 See infra Section V.C.2(b)(i) (discussing fees for consolidated market data).

consolidated market data product that contains all the elements of expanded core data because it will reduce information asymmetries between these market participants and market participants that subscribe to proprietary DOB feeds.[1938] Additionally, if a consolidated market data product containing all data elements is offered at reduced latency, then some market participants that currently rely on aggregated proprietary DOB feeds may use it as a substitute for proprietary feeds.[1939] Furthermore, there are likely market participants that may only benefit by taking subcomponents of expanded core data or products that competing consolidators offer that may be derived from the expanded content of core data, such as products that detail the best-priced odd-lot quotes or DOB imbalance measures. Therefore, to the extent that the individual components of expanded core data are less expensive than equivalent data from proprietary feeds,[1940] there will be demand for competing consolidators to also offer consolidated market data products that contain a subset of consolidated market data. Even if market participants do not directly benefit from any of the expanded content of core data, they may benefit indirectly if the broker-dealers that handle their orders subscribe to the expanded content.[1941] While the Commission believes

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1938 See infra Sections V.C.1(c)(i), V.C.1(c)(ii), and V.C.1(c)(iii). Commenters agreed that core data that included odd-lot information, auction information, and five levels of depth would be useful to market participants. See, e.g., Better Markets Letter at 3 ("These information taken together amount would fill a significant gap that currently exists in the SIP data."); ICI Letter at 4; State Street Letter at 2-3.

1939 See infra Section V.C.4(a). Commenters agreed that the additional information contained in expanded core data would make consolidated market data a viable alternative to proprietary DOB feeds. See, e.g., SIFMA Letter at 7; T Rowe Price Letter at 2; Clearpool Letter at 11.

1940 See infra Section V.C.2(b)(ii).

1941 See supra Sections II.A and II.C.2(a). See also infra Sections V.C.1(c)(i), V.C.1(c)(ii), and V.C.1(c)(iii).

there will be demand for the expanded content of core data, the Commission remains unable to quantify the number of market participants who will subscribe to the expanded content of core data because it does not have information on the number of market participants that would subscribe to proprietary DOB feeds, but do not do so because of the cost, or information on the number of market participants that currently subscribe to proprietary DOB feeds but might switch to expanded core data if the cost is lower.[1942]

Because competing consolidators will not be required to offer a consolidated market data product that contains all of the data elements of consolidated market data,[1943] there is a risk that a consolidated market data product containing all of the data elements of expanded core data will not be offered by any competing consolidator. The Commission believes this risk is low because there is likely to be sufficient demand for such a product from market participants. As discussed above in this section, because the fees for a consolidated market data product containing all of the data elements of core data are likely to be lower than fees for equivalent data from proprietary feeds today,[1944] the Commission believes that there will be demand from market participants for a consolidated market data product containing all of the elements of expanded core data.[1945] Because there will be demand for the data and because the competing consolidator market is subject to competitive forces, the Commission believes that one or more competing

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1942 See infra Sections V.C.2(b), V.C.4(a).

1943 See supra III.C.8(a).

1944 See infra Section V.C.2(b)(i) (discussing fees for consolidated market data).

1945 See supra note 1936 and accompanying text.

consolidators will be incentivized to offer a consolidated market product containing all of the data elements.

Commenters stated that expanding the content of core data would provide no benefit to retail investors.[1946] Commenters stated that depth of book and auction data is not useful for most retail investors and is likely to cause confusion.[1947] The Commission disagrees with these commenters. The Commission acknowledges that many retail investors may not directly view the entire content of expanded core data, but believes that retail investors will benefit from the expansion of the content of core data. Competing consolidators could offer customized products derived from the expanded content of core data that retail brokers may be able offer to their clients, who may be able to utilize the data to achieve some of the benefits discussed below without the retail broker taking in the additional message traffic from the full content of expanded core data. For example, competing consolidators could offer measures summarizing DOB or auction imbalances, or a feed that gives information on the best priced odd-lot quotes. Additionally, the Commission believes, as suggested by commenters, that retail brokers may allow some sophisticated retail investors to directly utilize the expanded content of core data and realize the benefits discussed below.[1948] Furthermore, retail investors may indirectly benefit if their executing broker-dealer uses expanded core data and did not previously receive this information from proprietary feeds. Additionally, retail investors may also indirectly benefit from other market participants utilizing expanded core data because they would be better able to

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1946 See, e.g., Nasdaq Letter IV at 33, 38; TD Ameritrade Letter at 2, 15.

1947 See, e.g., Nasdaq Letter IV at 33; TD Ameritrade Letter at 5.

1948 See, e.g., Schwab Letter at 1, 3.

observe and interact with retail investor orders, possibly leading to additional gains from trade.[1949]

(i) Effects of Addition of Information on Odd-Lot Quotes at and Inside the NBBO

This section discusses the economic effects of expanding the content of core data to include information on odd-lot quotes that are priced at or more aggressively than the NBBO to the definition of core data.[1950] For market participants who currently do not receive information on odd-lot quotes and choose to receive this aspect of expanded core data,[1951] the Commission generally believes that the economic effects will be similar to many of the effects discussed above regarding including smaller sized odd-lot quotes in the definition of a round lot.[1952] However, these benefits may be greater because these market participants will receive significantly more information on odd-lot quotes, since they will receive aggregated information on all odd-lot quotes priced better at or better than the NBBO for all NMS stocks, rather than just information on the smaller subset of quotes that will be included in the new round lot definition for stocks priced greater than $250.[1953] More specifically, the inclusion of odd-lot quote information in core data will improve transparency and reduce information asymmetry between market participants who already receive this information through proprietary DOB feeds and

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1949 See infra Sections V.C.1(c)(i), V.C.1(c)(ii), V.C.1(c)(iii).

1950 See supra Section II.C.2(b).

1951 Market participants may choose not to subscribe to this element, as well as other aspects of expanded core data. See infra Section V.C.1(c)(iv).

1952 See supra Section V.C.1(b)(ii).

1953 See supra Section V.C.1(b).

market participants who choose to subscribe to this aspect of core data and previously did not receive this information.[1954] This could potentially lead to these market participants being able to reduce their execution costs, make more informed trading decisions, facilitate best execution, as well as realize gains from trade. Including odd-lot quotes in core data may also cause changes in order flow to exchanges and off-exchange trading venues, as well as improvements in price efficiency. It may also benefit some market participants that currently subscribe to proprietary DOB feeds to receive data on odd-lot quotes because it may allow these market participants to receive this information through expanded core data, potentially at lower cost.[1955] However there may also be costs to market participants who choose to receive this data because they may need to upgrade their infrastructure in order to handle the additional message traffic contained in the odd-lot information.[1956] There could also be costs to market participants who currently receive information about odd-lot quotes from proprietary feeds and benefit from existing information asymmetries.

The Commission recognizes that many market participants, including many retail brokers-dealers (and their clients), may choose not to receive all of the information on odd-lot quotes priced at or better than the NBBO that is contained in expanded core data.[1957] However, the Commission believes that there are some market participants that currently do not receive

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1954 One commenter stated that including all odd-lot quotes at prices better than the protected BBO in core data would provide investors with valuable information. See CBOE Letter at 15.

1955 See infra Sections V.C.2(b) and V.C.4(a).

1956 See infra Section V.C.1(c)(iv).

1957 See supra Section V.C.1(c).

information on odd-lot quotes but may choose to receive this information from expanded core data if it is available at a cheaper price than equivalent proprietary data.[1958] If these market participants subscribe to this element of core data, then the Commission believes they will receive many of the benefits (and incur many of the costs) discussed below. Even if market participants do not directly receive all of the odd-lot information in expanded core data, they could realize some of the benefits if competing consolidators offer products that are derived from or contain some of the odd-lot information in expanded core data. For example, competing consolidators could offer a product that only contains information on the best priced odd-lot on each exchange. Because such a product would not significantly increase message traffic compared to receiving all the odd-lot information in expanded core data, many market participants, including many retail broker-dealers (who may offer it to their clients), may be able to utilize such a product and gain additional information about odd-lot quotes that would allow them to lower their execution costs and potentially realize additional gains from trade. Even if market participants do not receive any additional information on odd-lot quotes contained in expanded core data, they could still benefit if the broker-dealers handling their orders use the information. If a broker-dealer previously did not have access to odd-lot information, then a broker-dealer receiving the additional information may help facilitate best execution of its clients' orders. Even if a broker-dealer previously received the data from proprietary feeds and now receives it from core data, customers of the broker-dealer may benefit if the broker-dealer indirectly passes on any cost savings from switching data sources to its clients.

Adding information on odd-lot quotes that are priced at or more aggressively than the NBBO to the definition of core data will significantly increase transparency for market

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[1958] See infra Section V.C.2(b).

participants that do not currently receive information on odd-lot quotes, such as market participants that rely exclusively on SIP data, and choose to receive this element of expanded core data. Even though the new round lot definition would expand information on odd-lots that may be priced better than the current NBBO in some stocks,[1959] most stocks would not be affected by the new round lot definition.[1960] Additionally, the analysis in Table 1 shows that a substantial amount of odd-lot transaction volume in stocks above $250 would not be included in the new round lot definition. The addition of odd-lot information to expanded core data will make information on these additional odd-lot quotes that are priced at or better than the NBBO available to market participants who previously did not observe this information and who will choose to subscribe to this element of expanded core data. This would reduce information asymmetry between these market participants and market participants who currently receive this information through proprietary DOB feeds.

Market participants who choose to receive the odd-lot quotes from expanded core data and currently do not receive this information could realize a benefit from additional gains from trade. Some of these market participants may have traded with a price-improving odd-lot quote but did not because they cannot see information on odd-lot quotes. Under the final amendments, these market participants would be able to see these quotes if they receive odd-lot information from expanded core data, and make a decision about whether to trade based on this newly visible trading interest.[1961] This may benefit these market participants or their clients because they will

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1959 See supra Section V.C.1(b)(ii).

1960 The Commission estimates that approximately 98.5% of NMS stocks will have a round lot size of 100 shares. See supra Table 4.

1961 One commenter agreed that displaying odd-lot information would reveal greater liquidity in a stock. See RBC Letter at 5.